

Optimization of multi-stage constant current charging pattern based on Taguchi method for Li-Ion battery

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HIGHLIGHTS

- Taguchi method reduces the experimental cost and avoids the modeling of battery.
- It is universal to add subjectivity to choice of optimal solutions rather than objectives.
- The influences of charging currents are revealed to choose the optimal pattern.
- Charging efficiency is improved without reducing the time and capacity performance.

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ABSTRACT

Due to the complexity of characteristics, the charging performance of Li-ion batteries needs to be further improved. In this paper, Taguchi method is employed to search an optimal charging pattern for 5-stage constant-current charging strategy. The charging capacity, efficiency and time are analyzed as quality functions simultaneously, and the influences of charging currents at each stage on each quality function are revealed. This universal method provides a reasonable basis for the selection of the optimal currents. By reasonably updating the currents at each stage, a broader range is searched to make experimental results more representative. Compared with the constant current-constant voltage method, the obtained charging pattern improves the charging efficiency by 0.6–0.9%, and the temperature rise of the battery is reduced by about 2 °C. Compared with the charging pattern obtained by optimizing the charging time and capacity, the charging pattern obtained in this paper improved the charging efficiency by 2.8%, the temperature rise is reduced by 9.3 °C, and the charging capacity is basically the same.

1. Introduction

With the reduction of fossil energy and the increasing environmental problems, the development of renewable energy, such as electric transportation [1,2] and hybrid micro-grids [3], has received more and more attention. Energy storage is a necessity for the renewable energy generation [4], distributed network [5,6] and other occasions [7]. Li-ion batteries have emerged as an important choice for these applications, due to its long cycle life [8], high power density [9], low self-discharge rate and the absence of memory effect [10].

At present, there are many researches on the topic of the charging method. Charging control generally relies on accurate state of charge (SOC) estimation of Li-ion batteries [11]. Constant current-constant voltage (CC-CV) method avoids damage to the battery caused by large

current in the initial stage of CV and the end of CC mode [12]. Due to the wide application, the influence of current applied in CC stage and the charge cut-off voltage on the performance have also been studied [13]. The researchers have proposed many other charging strategies to improve the charging performance, such as considering polarization effects [14] and thermal characteristics [15], pulse and smart charging technology [16]. Inserting a pause between pulse currents is considered to eliminate polarization and improve charging performance [17]. A smart charging strategy requires charging parameters to change with the SOC and the state of health (SOH) [18]. The nature of the quality of the charging strategy should be analyzed from an electrochemical perspective. Therefore, many charging methods are studied based on electrochemical models. Such as the polarization of the battery [19,20], and the diffusion of ions [21,22]. The above charging strategy relies

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either on a special charger or on a complex battery model, while the CC-CV method as the most simply strategy is still the mainstream adopted by the commercial charger. Thus, the researchers proposed the multi-stage constant current (MS-CC) charging to increase the flexibility of charging strategy on the basis of CC-CV method without causing an additional burden on the commercial charger [23]. Therefore, MS-CC method is deeply studied in this paper.

A better strategy should maximize the charging capacity and efficiency, while minimize the charging time and temperature. Some of these indicators are contradictory and cannot be optimal at the same time. The solution of the optimal current pattern in MS-CC method can be regarded as an optimization problem. Ref. [24] obtained the optimal current pattern through orthogonal experiment to maximize the charging capacity. By weighting, the charging capacity and time are simultaneously optimized based on the Taguchi method, and the weights of both targets are equal to 0.5 [25]. Similarly, the discharge capacity and charging time were weighted in the same way, after the first set of orthogonal experiments, the evolutionary idea of the particle swarm optimization (PSO) was used in the iteration of the experiment [26]. Ref. [27] takes discharging capacity and charging time as optimal targets and the particle swarm algorithm is used. In this process, the fuzzy logic controller is used to make the optimal decision, which approximates specialist intuitiveness to make the optimal results more reasonable than weighting and priority processing. In addition to the terminal voltage, the charging strategy is also segmented based on the SOC [28]. Using intelligent algorithms to solve optimization problems relies on accurate battery models. There are certain limitations in the optimal solution obtained by weighting and adding priorities to deal with different objectives, and the influence of control variables on objectives is difficult to reveal intuitively.

In view of the aforementioned study, this paper provides an in-depth analysis of MS-CC strategy with Li-ion battery and the charging process is divided into five stages. For this multi-factor and multi-level problem, an optimization method based on the Taguchi method is carried out to search for the optimal charging pattern, and the orthogonal experiments greatly reduce the experimental cost compared to the exhaustive method. The selection of orthogonal array and the update of orthogonal experiments are determined by the actual situation. Taguchi method is based on experimental analysis to avoid the complicated modeling of Li-ion battery. Charging efficiency, time and capacity are analyzed as the optimal objectives simultaneously, and these objectives are parallel in the analysis process. Instead of turn the multi-objective into a single by weighting, this analysis adds subjective factors to the choice of the optimal solution rather than the objective function. Therefore, the influences of current at every stage on each objective are revealed, and the choice of optimal charging pattern is based on the characteristics of the factor effect curves. Moreover, for different charging occasions, the obtained factor effect is also effective and can be a versatile basis and inspiration for the optimal decision. Thus, this idea is suited for the optimization of MS-CC charging strategy for Li-ion batteries.

This paper is organized into 6 sections. Section 2 introduces the basic concept of the Taguchi method. Section 3 is the formulation of 5-stage constant current (5S-CC) charging method. On this basis, Section 4 introduces the experimental design. Section 5 provides experimental analysis. Section 6 is the conclusion of this paper. Commercially available 18,650 Li-ion battery was used in experiments. The rated capacity of the battery is 3150 mAh, and the maximum allowable charging current is 2 C (C means the rate, which can define the current corresponding to complete charging or discharging the battery in one hour). The maximum allowable voltage is 4.2 V, the minimum allowable voltage is 2.5 V and the rated voltage is 3.7 V.

2. Basic principle of the Taguchi method

The Taguchi method is one of the design of experiment (DOE) in quality engineering [17,25]. Two important tools of the Taguchi

Table 1
Typical orthogonal experiment.

Expt No.	Factors			Quality			S/N
	F_1		F_C	Y_1		Y_n	
1	L_{11}		L_{1C}	Y_{11}		Y_{1n}	S/N_1
2	L_{21}		L_{2C}	Y_{21}		Y_{2n}	S/N_2
:	:		:	:		:	:
:	:		:	:		:	:
A	L_{A1}		L_{AC}	Y_{A1}		Y_{An}	S/N_A

method are the orthogonal experiment and the signal-to-noise ratio (S/N).

2.1. Orthogonal experiment

Orthogonal experiment uses orthogonal array (OA) to design and analyze the multi-factor and multi-level experiments. A typical orthogonal experiment is shown in Table 1, where F is the system factor, L is the element of the OA and Y is the system output, which is the measured value of the quality [26].

The OA has two basic properties. (1) Each level in any column occurs and the number of occurrences is equal. (2) Possible combinations of different levels between any two columns occur and the number of occurrences is equal. These properties enable orthogonal experiments to be evenly distributed. It eliminates the interference between the factors, and make it possible to analyze the influence of different levels with the same factor on the quality. In general, equal-level OA is expressed in the form of $L_A(B^C)$, where L represents the orthogonal design, and A , B , C represent the number of experiments, number of levels of each factor and number of factors, respectively [25].

2.2. Signal to noise ratio

The factor effect refers to the influence of control factor on the quality function. In the Taguchi method, generally, the S/N is adopted as the indicator of quality. It can be derived as follows:

$$S/N_a = \begin{cases} -10 \lg \left(\frac{1}{n} \sum_{i=1}^n Y_{ai}^2 \right) & \text{The smaller the better} \\ -10 \lg \left(\frac{1}{n} \sum_{i=1}^n Y_{ai}^{-2} \right) & \text{The larger the better } (1 \leq a \leq A) \\ -10 \lg \left(S_a^2 \left(\frac{1}{n} \sum_{i=1}^n Y_{ai} \right)^{-2} \right) & \text{Nominal is the best} \end{cases} \quad (1)$$

where i means the i -th sample and S_a^2 is the quality variance in the a -th experiment. It can be derived as follows:

$$S_a^2 = \frac{1}{n} \sum_{i=1}^n \left(Y_{ai} - \frac{1}{n} \sum_{i=1}^n Y_{ai} \right)^2 \quad (2)$$

The average S/N is employed to determine the optimal level of each control factor. It can be derived as follows:

$$\overline{S/N}_{lbd} = \frac{1}{m_{bd}} \sum_{a=1}^A (\partial_{bd}(L_{ac}, l_{bd}) \cdot S/N_a) \quad (3)$$

with

$$\begin{cases} m_{bd} = \sum_{a=1}^A \partial_{bd}(L_{ac}, l_{bd}) \\ \partial_{bd}(L_{ac}, l_{bd}) = \begin{cases} 1 & (L_{ac} = l_{bd}) \\ 0 & (L_{ac} \neq l_{bd}) \end{cases} \end{cases} \quad (4)$$

where L_{ac} is the element of the OA at the a -th row and the c -th column, and l_{bc} is the b -th level of the d -th factor.

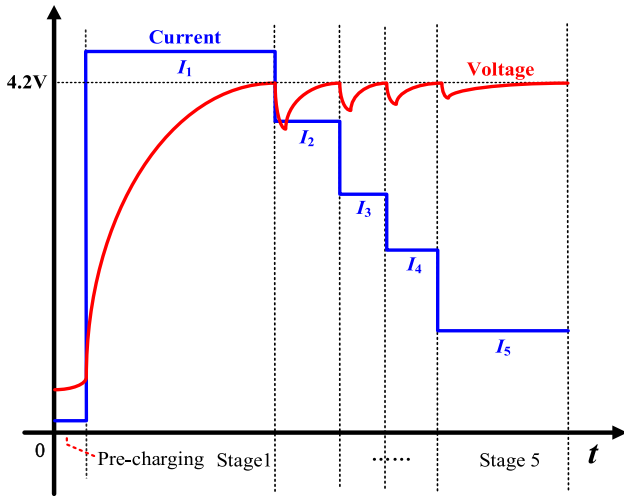


Fig. 1. The 5-stage constant current charging algorithm.

3. Formulation of 5-stage constant current charging method

CC-CV method is widely used because of its simplicity and effectiveness. However, it is not flexible, and the MS-CC is an improvement of the constant-voltage (CV) stage in CC-CV method. It not only has the advantages of CC-CV method, but increases its flexibility.

Fig. 1 illustrates the charging curve for 5S-CC method. The battery terminal voltage continues to increase at each stage, when the voltage reaches the preset limitation, the charging current jumps to the next value until the last stage. The current of each stage of the 5S-CC method is gradually reduced, which is expressed as follows:

$$I_{\max} \geq I_1 > I_2 > I_3 > I_4 > I_5 > 0 \quad (5)$$

where $I_{\max}$ is the maximum acceptable charging current of the battery.

Usually, a large rate results a shorter charging time but will reduce the charging capacity due to the internal resistance and polarization of the battery. It suggests that the charging time and capacity contradict with each other. In this paper, the charging time and capacity are chosen as the quality function. They are expressed as follows:

$$Y_{soc}(a) = \frac{Q(a)}{Q_{rate}} \quad (6)$$

$$Y_{time}(a) = Y_{soc-\min} + \frac{T_{\max} - T(a)}{T_{\max} - T_{\min}} (Y_{soc-\max} - Y_{soc-\min}) \quad (7)$$

where Q is the charging capacity and Q_{rate} is the rated capacity of the battery, T represents charging time. Their maximum and minimum values are marked with corresponding subscripts.

The equivalent circuit of the Li-ion battery is shown in Fig. 2, where V_o is the terminal voltage, the open-circuit voltage (OCV) of the battery is represented by a voltage source, R_Ω is the internal resistance and Z_P is the polarization impedance. V_R and V_P is the voltage of R_Ω and Z_P ,

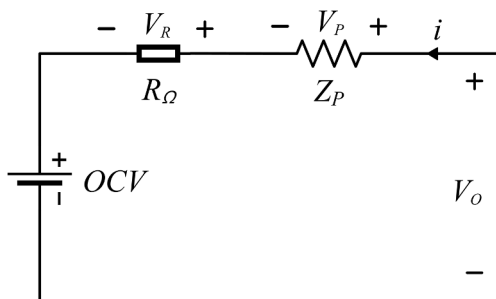


Fig. 2. Equivalent circuit of Li-ion battery.

respectively. Two assumption can be made: (1) the polarization voltages during the charging and discharging process with the same amplitude current should be the same magnitude and opposite polarity; (2) the internal resistance of the battery is constant. Thus, V_o are expressed as

$$V_{o-ch} = OCV + V_R + V_P \quad (8)$$

$$V_{o-disch} = OCV - V_R - V_P \quad (9)$$

where V_{o-ch} represents the terminal voltage of the battery during the charging process, and $V_{o-disch}$ represents the terminal voltage during the discharging process.

The OCV is related to SOC, which is obtained by calculating the average of the charging and the discharging curve at 0.05 C current. So, the OCV can be obtained as follows:

$$OCV(SOC) = \frac{V_{o-ch}(SOC) + V_{o-disch}(SOC)}{2} \quad (10)$$

The curve of the OCV-SOC is shown in Fig. 3. During the charging process, the actual energy obtained by the battery is reflected in the OCV, and the energy output from the charger is reflected in the terminal voltage V_o . According to the battery thermal model, the heat generation is caused by the energy loss and the reversible reaction heat, where the energy loss is the major reason [29]. Reducing the charging loss not only saves energy but alleviates temperature rise of the battery. Therefore, charging efficiency is also considered as one of the quality functions, which can be derived as follows:

$$\eta(a) = \frac{\int_0^t OCV(SOC) i_a(t) dt}{\int_0^t V_{o-ch-a}(t) i_a(t) dt} \quad (11)$$

where V_{o-ch-a} and i_a represents the terminal voltage and charging current of the battery during the a -th experiment respectively. Thus, the efficiency is standardized as follows:

$$Y_\eta(a) = Y_{soc-\min} + \frac{\eta(a) - \eta_{\min}}{\eta_{\max} - \eta_{\min}} (Y_{soc-\max} - Y_{soc-\min}) \quad (12)$$

where the label of the subscript is similar to that in Eq. (7), and the maximum and minimum values in Eqs. (7) and (12) are taken from the current orthogonal experiment.

4. Experimental design based on Taguchi orthogonal arrays

Taguchi method is effectively used to solve the multi-factor optimization. Firstly, the maximum charging current is 2 C, which is divided into 20 parts at intervals of 0.1 C. Therefore, the current of each stage contains 4 levels in the 5S-CC method and the control factors of

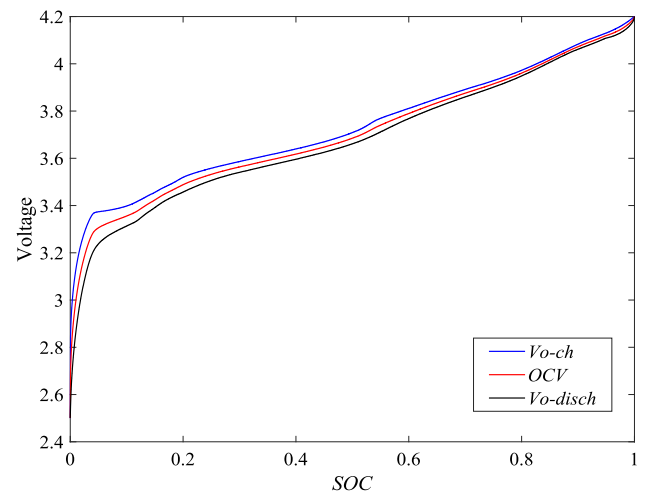


Fig. 3. Waveforms of OCV-SOC for Li-ion battery.

Table 2
Current candidates for the first orthogonal experiments.

Level (A)	Factors				
	I_1	I_2	I_3	I_4	I_5
1	5.355	4.095	2.835	1.575	0.315
2	5.67	4.41	3.15	1.89	0.63
3	5.985	4.725	3.465	2.205	0.945
4	6.3	5.04	3.78	2.52	1.26

Taguchi experiment and their levels are shown in Table 2. Since the current is limited by Eq. (5), the current candidates in Table 2 are not complete. For example, when the optimal current of the first stage is level 4, the current of the second stage can be increased based on the present. Therefore, it is necessary to update Table 2 based on the current experimental results to carry out the next orthogonal experiments.

Secondly, the appropriate OA should be selected based on the minimum experimental cost. According to the control factor in Table 2, there are five factors corresponding to the currents of five charging stages, and each factor contains four levels. Therefore, the OA with five-factor and four-level is selected to be denoted as $L_{16}(4^5)$ as shown in Table 3, which needs to be changed with the update of Table 2 if necessary.

Thirdly, orthogonal experiments should be carried out. Each charging pattern is programmed into the test device. To minimize the impact on battery performance, the same discharge method is adopted. All experiments were carried out with 22 °C ambient temperature. The experimental steps are as follows:

Step 1: The discharge of the battery is carried out by CC method with 0.2 C current. When the voltage of battery reaches 2.5 V, the discharge is terminated.

Step 2: The battery is left to stand for 1 h. One purpose is to make the battery temperature the same as the ambient, and the other is to make the battery terminal voltage stable.

Step 3: Pre-charge the battery at a current of 1/20 C when the SOC is less than 3%. It not only improves the battery's charging performance, but also provides a uniform initial value for all orthogonal experiments.

Step 4: Charge the battery by a pre-set current pattern according to Tables 2 and 3. Charging limit voltage is 4.2 V, and the experimental data is recorded.

Step 5: The battery is left to stand for 1 h. Go back to step 1 to discharge the battery. Then a cycle of charge and discharge is

Table 3
 $L_{16}(4^5)$ orthogonal array.

No.	Factors				
	F_1	F_2	F_3	F_4	F_5
1	1	1	1	1	1
2	1	2	2	2	2
3	1	3	3	3	3
4	1	4	4	4	4
5	2	1	2	3	4
6	2	2	1	4	3
7	2	3	4	1	2
8	2	4	3	2	1
9	3	1	3	4	2
10	3	2	4	3	1
11	3	3	1	2	4
12	3	4	2	1	3
13	4	1	4	2	3
14	4	2	3	1	4
15	4	3	2	4	1
16	4	4	1	3	2

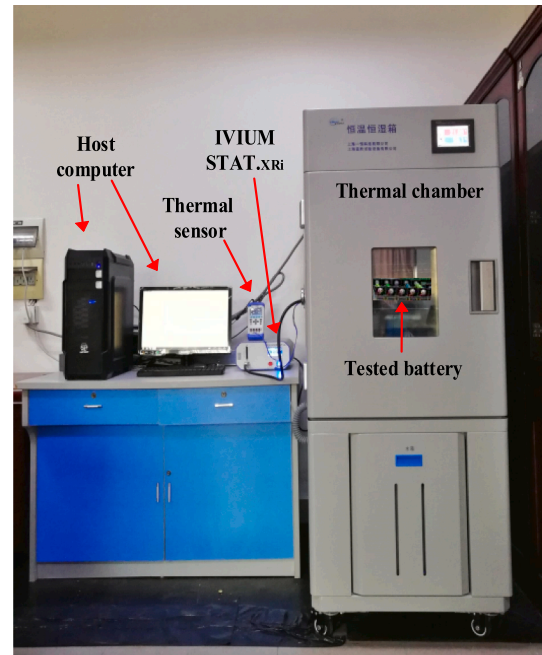


Fig. 4. Configuration of test platform.

completed. This process ends when all the experiments in Table 3 are performed.

The factor effect of the current at each stage is calculated, which reflects influence of the current at each stage on each objective. The choice of the optimal pattern should be constructed according to the various application. If further experiments are needed, Table 2 needs to be updated based on the experimental results and the current optimal pattern. Therefore, the experimental design returns to the first step until the optimal pattern converges.

5. Analysis of experimental results

The charging/discharging instrument used in this paper is the ultra-high resolution electrochemical workstation produced by IVIUM corporation. The temperature tester from the Applent Instruments is used to monitor the temperature of the battery. The test platform is shown in Fig. 4.

5.1. The first orthogonal experimental results

The Taguchi orthogonal experiment is carried out in an ordered manner according to the experimental design scheme in Section 4. The charging capacity and time can be directly measured, and the charging efficiency can be calculated based on Eq. (11). Furthermore, the quality functions are calculated according to Eqs. (6), (7) and (12) respectively, and then their S/N can be obtained by Eq. (1). The results of the experiments mentioned above are shown in Table 4.

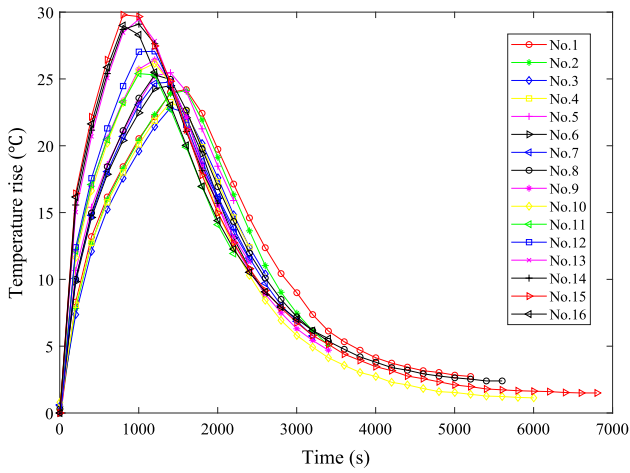
According to Table 4, the No.14 experiment with the shortest time (i.e. 2200 s) was accompanied by the minimum charging capacity and efficiency, while the No.15 with the longest time (i.e. 6980 s) was also very low in charging efficiency and capacity. Fig. 5 is the curve of the temperature rise of the battery surface in the first orthogonal experiment. The temperature rise are significantly different, and the speed of the temperature rise mainly depends on the first stage current, and the temperature decreases correspondingly with the current in the later stage. Therefore, it is necessary to optimize the charging currents.

Then the average S/N ratio of each level of current in each stage can be calculated by Eq. (3), which shown in Tables 5–7. Fig. 6 is a factor

Table 4

The first Taguchi orthogonal experiment results.

No Exp.	Time (s)	Capacity (C)	Efficiency (%)	Y_{time}	Y_{SOC}	Y_{η}	$S/N_{Y_{time}}$	$S/N_{Y_{SOC}}$	$S/N_{Y_{\eta}}$
1	5280	10828.29	0.9245	0.85852439	0.95538115	0.95538115	-1.32494719	-0.39646659	-0.39646659
2	3420	10456.16	0.9225	0.91701581	0.92254808	0.94519019	-0.75246349	-0.70021976	-0.48961585
3	2796	10198.51	0.9205	0.93663874	0.89981559	0.93499923	-0.56855764	-0.91692964	-0.58377489
4	2442	9928.287	0.9183	0.94777098	0.87597379	0.92378917	-0.46593186	-1.15017770	-0.68854259
5	2308	9885.625	0.914	0.95198487	0.87220972	0.90187861	-0.42739900	-1.18758152	-0.89703821
6	2762	10061.19	0.9157	0.93770794	0.88769984	0.91054093	-0.55864812	-1.03467715	-0.81401054
7	3338	10297.65	0.9168	0.91959446	0.90856273	0.91614595	-0.72807299	-0.83290163	-0.76070659
8	5786	10563.48	0.9172	0.84261221	0.93201694	0.91818415	-1.48744500	-0.61152387	-0.74140416
9	3584	10195.39	0.9129	0.91185850	0.89954032	0.89627358	-0.80145092	-0.91958730	-0.95118804
10	6138	10473.35	0.9146	0.83154287	0.92406476	0.90493590	-1.60230712	-0.68595182	-0.86764362
11	2202	9520.932	0.9059	0.95531826	0.84003282	0.86060522	-0.39703842	-1.51407489	-1.30392040
12	2560	9674.526	0.904	0.94406023	0.85358443	0.85092381	-0.50000590	-1.37507024	-1.40218642
13	2562	9510.326	0.8992	0.94399734	0.83909705	0.82646551	-0.50058458	-1.52375608	-1.65550529
14	2200	9124.601	0.895	0.95538115	0.80506449	0.80506449	-0.39646659	-1.88338651	-1.88338651
15	6980	9916.411	0.9	0.80506449	0.87492597	0.83054189	-1.88338651	-1.16057379	-1.61276910
16	3502	9410.733	0.8972	0.91443716	0.83030995	0.81627455	-0.7769226	-1.61519512	-1.76327485

**Fig. 5.** Waveforms of the temperature rise in the first orthogonal experimental.**Table 5**Average S/N ratio of each current level for Y_{SOC} .

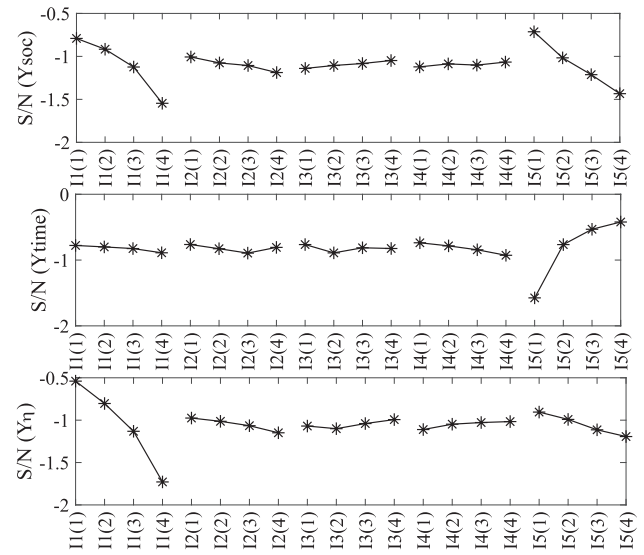
$\overline{S/N_{Y_{SOC}}}$	I_1	I_2	I_3	I_4	I_5
1	-0.790948	-1.006847	-1.140103	-1.121956	-0.713629
2	-0.916671	-1.076058	-1.105861	-1.087393	-1.016975
3	-1.123671	-1.106119	-1.082856	-1.101414	-1.212608
4	-1.545727	-1.187991	-1.048196	-1.066253	-1.433805

Table 6Average S/N ratio of each current level for Y_{time} .

$\overline{S/N_{Y_{time}}}$	I_1	I_2	I_3	I_4	I_5
1	-0.777975	-0.763595	-0.764389	-0.737373	-1.574521
2	-0.800391	-0.827471	-0.890813	-0.784382	-0.764727
3	-0.825200	-0.894263	-0.813480	-0.843796	-0.531949
4	-0.889340	-0.807576	-0.824224	-0.927354	-0.421708

Table 7Average S/N ratio of each current level for Y_{η} .

$\overline{S/N_{Y_{\eta}}}$	I_1	I_2	I_3	I_4	I_5
1	-0.539599	-0.975049	-1.069418	-1.110686	-0.904570
2	-0.803289	-1.013664	-1.100402	-1.047611	-0.991196
3	-1.131234	-1.065292	-1.039938	-1.027932	-1.113869
4	-1.728733	-1.148852	-0.993099	-1.016627	-1.193221

**Fig. 6.** Average S/N of each current level for the first orthogonal experimental.

effect diagram with data from Tables 5–7. Fig. 7 is the mean value of charging capacity, time and efficiency of each current level. It can be seen from Fig. 5 that the influence of the current at each stage on the quality function is clear. For the charging capacity, the currents of the first and fifth stage have the greatest influence, and the capacity decreases as the current increases. For the charging time, only the current of the fifth stage has the greatest influence, and the charging time decreases as the current increases. The greatest influence on the efficiency is the current in the first stage, and the charging efficiency decreases as the current increases.

Significant factors can be used to adjust the quality function to the optimal, non-significant factors may be used to reduce the cost of experimentation. For the three quality functions, the maximum of the average S/N ratio of I_1 and I_2 appears at level 1, so level 1 is selected as the optimal current in these two stage. The fifth stage current as the only significant factor for the charging time can be used to optimize the charging speed, but a higher current leads to a shorter charging time and a lower charging capacity and efficiency. Therefore, the value of the current in the fifth stage needs to be weighed. Here, level 2 is selected as the optimal value because the charging time corresponding to level 1 is too long and exceeds 6000 s according to Fig. 6, and levels 3 and 4 significantly reduce the charging capacity and with a lower efficiency. According to the above analysis, besides the current of the fifth stage, the charging capacity and efficiency can be optimized by the

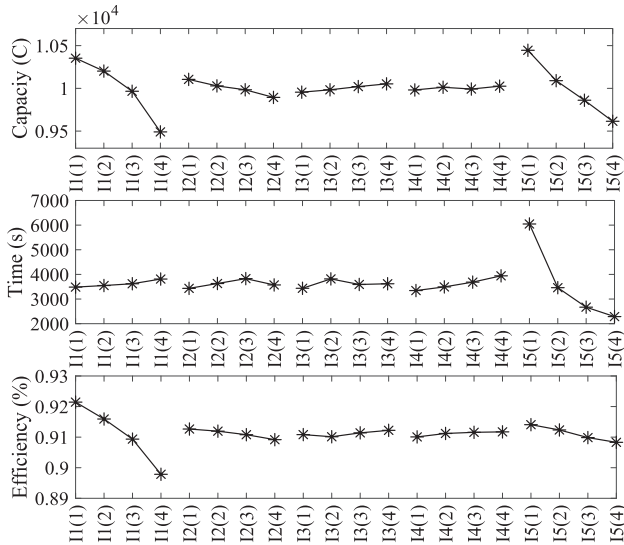


Fig. 7. Mean values of charging capacity, time and efficiency of each current level for the first orthogonal experiment.

current in the first stage. The current of the fifth stage is weighed, which is the only significant factor for the charging time. Thus, the charging time is difficult to further adjust. Therefore, as non-significant factors, levels 1 of the stage 3 and 4 are selected as the optimal currents on the basis of shorter time. In summary, the optimal charging current pattern obtained by the first orthogonal experiment is the combination ($I_1(1)$, $I_2(1)$, $I_3(1)$, $I_4(1)$, $I_5(2)$). These factor levels are denoted in Table 2 using bold characters.

5.2. The second orthogonal experimental results

According to Section 4, it is necessary to update Table 2 based on the experimental results. The average S/N ratio of the first and second stages decreases with the increase of current in Fig. 6, reducing the current may result in a better charging performance. According to the optimal current pattern in Table 2, the reduction of current in each stage is determined. Update the current values of each stage for second orthogonal experiment as shown in Table 8. The currents of stages 1, 2 and 3 can be reduced by 0.3 C based on the current optimum value, the current of stage 4 can be reduced by 0.2 C, and the current of stage 5 is fixed at 0.2 C. Here the dummy level technology is applied to handle the empty levels, which make the current in each stage still have 4 levels. As shown in Table 8, the levels 1 and 2 of the fourth stage current are 0.3 C (i.e., 0.945 A) in order to obtain more accurate experimental information of this boundary current. Each level of stage 5 is 0.2 C (i.e., 0.63 A). In the second experiment, the orthogonal Table 3 was still used. The results of the second experiment are shown in Table 9.

According to Table 9, the differences of each experiment are significantly reduced compared with the first experiment. In all experiments, the No. 4 has the longest charging time (i.e., 3720 s), and the No. 16 requires the shortest charging time (i.e., 3280 s), but it corresponds to the minimum charging capacity and efficiency. The difference between

Table 8
Current values for the second orthogonal experiments.

Level (A)	Factors				
	I_1	I_2	I_3	I_4	I_5
1	4.41	3.15	1.89	0.945	0.63
2	4.725	3.465	2.205	0.945	0.63
3	5.04	3.78	2.52	1.26	0.63
4	5.355	4.095	2.835	1.575	0.63

the maximum and minimum values of the charging capacity is about 220 C, and the difference in efficiency is slightly larger than 1%. The average S/N ratios of each level of current in each stage can be calculated as show in Figs. 8 and 9 is the average value of charging capacity, time and efficiency. As can be seen from Fig. 8, only the current in the first stage is a significant factor for each quality function. As the current increases, the charging time, capacity and efficiency will decrease. Although as a significant factor, the ability of the current of the first stage to adjust the quality function is limited. Here, the median value (i.e., 4.8825A/1.55 C) of the four levels in the first stage is used as the optimal charging current to weigh three different quality functions. The currents of stages 2, 3 and 4 are non-significant factors, and level 1 is selected as the optimal value, respectively. On the one hand, as can be seen from Figs. 8 and 9, other levels have no advantage over the level 1 for the quality functions. On the other hand, without degrading the charging performance, smaller currents can reduce the temperature rise and current stress of the charger and battery. Of course, the choice of the optimal currents is not unique, based on the experimental results, it needs chosen by reality. The optimal charging current pattern obtained by the second orthogonal experiment is (1.55 C, 1 C, 0.6 C, 0.3 C, 0.2 C). If necessary, the current in each stage can be further updated on the basis of the second orthogonal experiment. The trend of charging current influence the quality function at each stage has been reflected. Thus, further experiments have not been carried out.

5.3. Comparative experiments

In order to verify the performance of the obtained charging pattern, the first comparative experiment was carried out using CC-CV method and the obtained charging pattern. The current in the CC stage is 1.55 C (i.e., 4.8825 A), and the cut-off current of the CV stage is 0.1 C (i.e., 0.315 A). The CC-CV method and the obtained charging pattern were alternately tested three times to calculate the average of the results. The steps of the comparative experiment are consistent with the experiment design in the Section 4, except that the current pattern of orthogonal experiment is replaced by CC-CV strategy during CC-CV charging test. The second comparative experiment is compared with Ref. [25], and the quality function of Ref. [25] is as follows:

Normalization(i)

$$= \alpha \left(\frac{Q_i - 5step}{Q_i - CC/CV} \right) + \beta \left(\frac{Q_{min}}{Q_i - CC/CV} + \left(\frac{T_{max} - T_i}{T_{max} - T_{min}} \right) \left(\frac{Q_{max}}{Q_i - CC/CV} - \frac{Q_{min}}{Q_i - CC/CV} \right) \right) \quad (13)$$

where i refers to the i -th charging parrern, and Ci -cc/cv takes the rated capacity of the battery (3150 mAh/ 11,340C). The maximum and minimum values of capacity and time are from the two orthogonal experiments in this paper, which are 10828.29 C, 9124.601 C and 6980 s, 2200 s respectively. Both α and β are equal to 0.5. The current constraint also satisfies Eq. (5) and the maximum charging current is 2 C.

Unlike Ref. [25], the non-dominated sorting genetic algorithm (NSGA) is employed to solve the above optimization problem. The optimization process of the global optimal solution is shown in Fig. 10. The algorithm converges when it evolves to about the 130th generation, and the optimal current combination is (6.3 A, 4.3194 A, 2.3627 A, 1.1623 A, 0.6041 A). For convenience of distinction, this current combination is called pattern 2, and the optimal charging result obtained in this paper is called pattern 1.

The waveforms of battery temperature rise (represents Trise in the legend), terminal voltage and charging current in the comparative experiment are shown in Fig. 11. As show in Fig. 11(a), the CC stage of CC-CV method is consistent with the first stage of 5S-CC. The temperature rise reaches the maximum at the end of the first charging stage

Table 9
The second Taguchi orthogonal experiment results.

No Exp.	Time (s)	Capacity (C)	Efficiency (%)	Y_{time}	Y_{SOC}	Y_{η}	$S/N_{Y_{time}}$	$S/N_{Y_{SOC}}$	$S/N_{Y_{\eta}}$
1	3554	10651.735	0.9399	0.92809248	0.93980368	0.93813119	-0.64817484	-0.53925710	-0.55472849
2	3588	10639.8	0.9408	0.92663927	0.93875066	0.93963643	-0.66178594	-0.54899487	-0.540803
3	3624	10623.41	0.9409	0.92510057	0.93730457	0.93980368	-0.67622097	-0.56238530	-0.53925710
4	3730	10591.523	0.9394	0.92056996	0.93449117	0.93729494	-0.71886396	-0.58849589	-0.56247453
5	3474	10583.44	0.9364	0.93151181	0.93377801	0.93227744	-0.61623262	-0.59512712	-0.60909641
6	3596	10555.68	0.9364	0.92629734	0.93132874	0.93227744	-0.66499165	-0.61793984	-0.60909641
7	3586	10565.03	0.9372	0.92672475	0.93215369	0.93361544	-0.66098469	-0.61024947	-0.59663942
8	3520	10539.43	0.938	0.92954570	0.92989500	0.93495344	-0.63458504	-0.63132169	-0.58420027
9	3462	10546.57	0.9352	0.93202471	0.93052496	0.93027045	-0.61145141	-0.62543937	-0.62781546
10	3388	10516.61	0.9346	0.93518759	0.92788159	0.92926695	-0.58202525	-0.65014879	-0.63719014
11	3302	10504.83	0.9342	0.93886337	0.92684224	0.92859795	-0.54795206	-0.65988359	-0.64344554
12	3346	10467.2	0.9332	0.93698274	0.92352214	0.92692545	-0.56536816	-0.69105371	-0.65910380
13	3400	10451.09	0.9305	0.93467469	0.92210075	0.92240971	-0.58679029	-0.70443241	-0.70152264
14	3294	10448.06	0.9303	0.93920530	0.92183342	0.92207521	-0.54478925	-0.70695100	-0.70467304
15	3348	10440.48	0.9295	0.93689725	0.92116463	0.92073721	-0.56616063	-0.71325485	-0.71728605
16	3280	10433.74	0.9294	0.93980368	0.92056996	0.92056996	-0.53925710	-0.71886396	-0.71886396

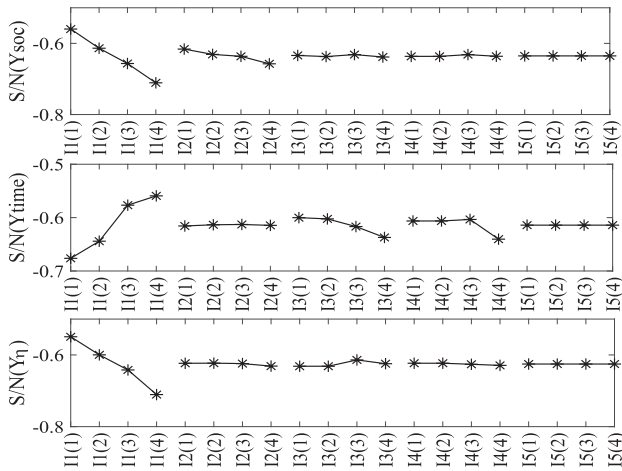


Fig. 8. Average S/N of each current level for the second orthogonal experimental.

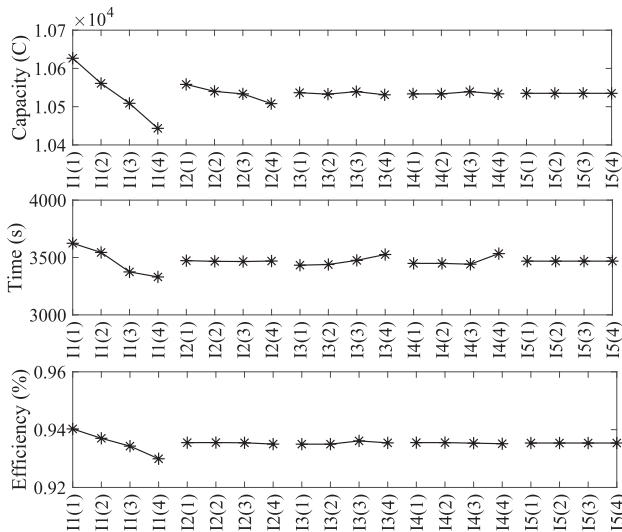


Fig. 9. Mean values of charging capacity, time and efficiency of each current level for the second orthogonal experiment.

for 5S-CC strategy, which is 19.2 °C, and then the battery temperature starts to drop. In the CV stage of CC-CV method, the current smoothly decreases until the cut-off current is reached. The temperature rise

reaches the maximum when the current drops to about 3.8 A, which is 20.3 °C. Thereafter, the battery temperature corresponding to the charging pattern 1 is about 2 °C lower than that of the CC-CV method due to the sudden drop in current at the end of the first charging stage. As shown in Fig. 11(b), the charging time required for pattern 1 and 2 are close, and the current of first stage in pattern 2 is much higher than that in the pattern 1. This resulted in the maximum temperature rise of pattern 2 to reach 28.5 °C, which is much higher than the 19.2 °C in pattern 1. From the current curves, the difference in charging capacity between the two patterns is not too large. This shows that increasing the charging current blindly does not shorten the entire charging process, and it also causes the battery to be severely heated and seriously degrades the charging performance, which poses a safety hazard to the application of the battery.

The results of the comparative experiment are shown in Fig. 12. Since the last current of the pattern 1 is 0.2 C, the experimental results of the CC-CV method are also presented when the current in the CV stage drops to 0.2 C, which are expressed by CC-CV-0.2C in Fig. 12. The cut-off current in the CV stage of CC-CV method is set to 0.1 C, and its experimental results are expressed as CC-CV-0.1C. The charging capacity of the CC-CV-0.2C is 92.3%, and the charging efficiency is 93%. Compared with the obtained charging pattern 1, these two indicators are reduced by 0.7% (ie. 79.3 C) and 0.9%, respectively. The stepwise reduction of the charging current in MSCC method actually reduces the charging speed, and in order to obtain more charging capacity, the charging time is inevitably increased. Thus, the charging time of the pattern 1 is 3253 s, which is 291 s higher than the CC-CV-0.2C.

As the charging cut-off current is reduced to 0.1 C, the charging capacity of CC-CV-0.1C is 94.1%, but the charging time is 3564 s, and the charging efficiency is 93.3%. The lower cut-off charging current can further increase the charging capacity, but the charging time is also greatly increased, and the charging efficiency is not significantly improved, which is still 0.6% lower than that in the pattern 1. This means that the charge cut-off current or the current in the last stage of MSCC strategy has a significant effect on the charging time and capacity, and it is not a significant factor for charging efficiency. The experimental results obtained using Eq. (13) as the quality function are shown as pattern 2 in Fig. 12. The charging capacity is basically the same as that of pattern 1, and the charging time is slightly increased, but the efficiency is the lowest at 91.1%. It shows that the charging process is a trade-off between multiple indicators. Compared with pattern 2, pattern 1 also takes into account the charging efficiency, which significantly improves the efficiency without reducing the charging time and capacity performance. Increasing the charging efficiency is also important for reducing the temperature rise. Thus, the obtained charging pattern is safer, especially in the packs where the batteries are

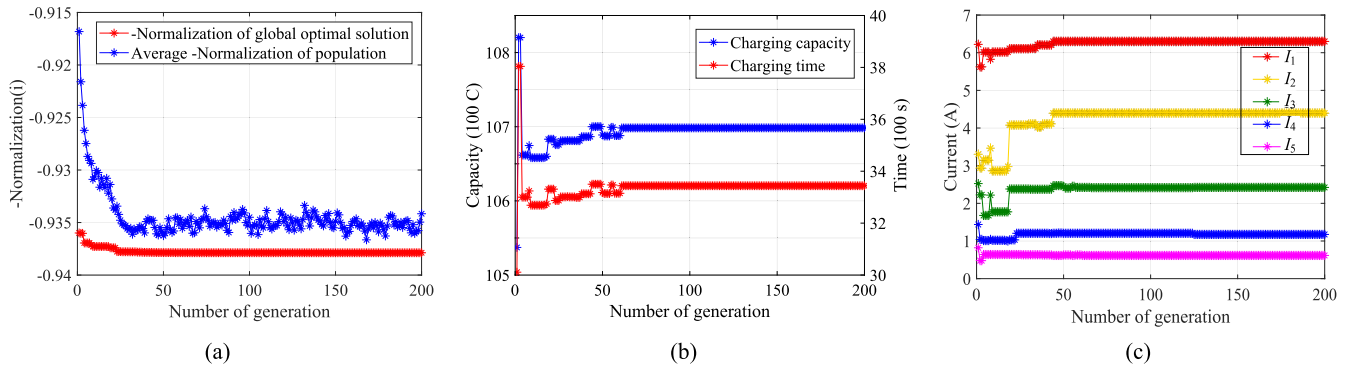


Fig. 10. Evolutionary process of the global optimal solution. (a). Normalization(i) and its average. (b) Charging capacity and time. (c) Charging current.

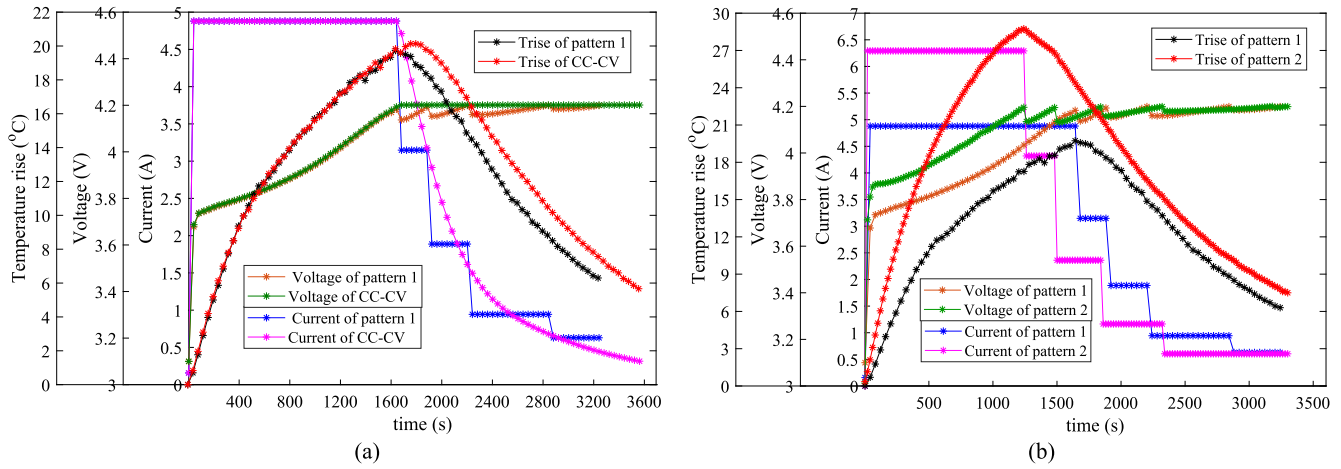


Fig. 11. Waveforms of the battery temperature rise, voltage and current in the comparative experiment. (a) Compared with CC-CV method. (b) Compared with charging pattern 2.

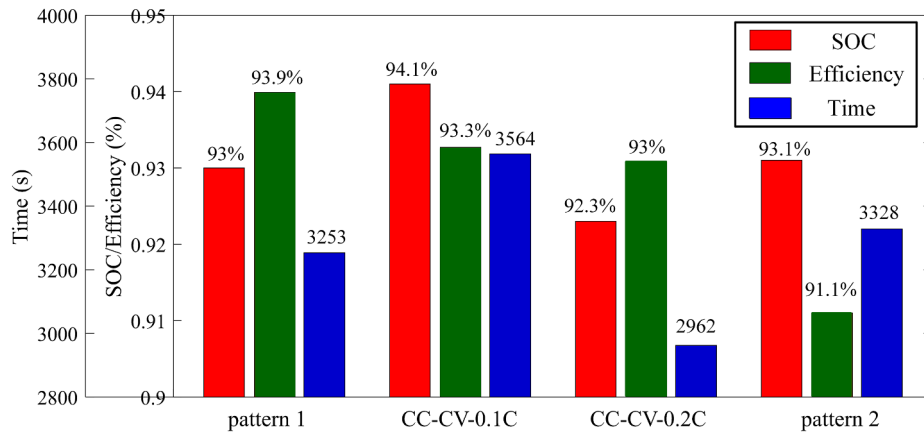


Fig. 12. Results of the comparative experiment.

packaged in large scale. Moreover, multiple charging indicators are not weighted into a single target, and the effect of charging currents on each indicator can be better revealed through the Taguchi experiments, which provides a reasonable basis for the selection of the optimal currents.

6. Conclusions

This paper provides an in-depth analysis of five-stage constant current charging strategy with Li-ion battery. An optimization based on the Taguchi method has been carry out. The major work of this paper includes as follows: (1) Multi-stage constant current strategy increases

the flexibility of charging process based on constant current-constant voltage method, and it can also be applied to mainstream charger on the market without increasing control complexity; (2) The charging capacity, time and efficiency are analyzed as quality functions simultaneously. The influences of charging current at each stage on each quality function are revealed intuitively. This universal method provides a reasonable basis for the selection of the optimal pattern; (3) The Taguchi method has a good performance for searching the optimal current pattern for this multi-factor and multi-level optimization. It greatly reduces the experimental cost compared to exhaustive experiments and avoids the complex modeling of Li-ion batteries; (4) By comparing with the constant current-constant voltage method and the

charging pattern obtained by optimizing the charging time and capacity, the charging pattern obtained in this paper greatly improves the charging efficiency, and significantly reduces the temperature rise of the battery during charging process.

Declaration of Competing Interest

We declare that we have no financial and personal relationships with other people or organizations that can inappropriately influence our work. There is no professional or other personal interest of any nature or kind in any product, service and/or company that could be construed as influencing the position presented in the manuscript submitted.

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